

Tran Hoang Kien

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ABOUT ME

Ho Chi Minh University of Technology - VNU HCM

Bachelor of Engineering in Electronics and Telecommunications (Fourth-year student)

2022 – Present

- A reliable and patient individual with strong problem-solving and planning skills.
- Passionate about electronics and embedded systems.
- Seeking opportunities to develop expertise in embedded programming.

SKILL

Technical Skills: C, MCU drivers (UART, SPI, I2C, CAN), Real-Time OS (FreeRTOS), Hardware design, USB stack, TCP/IP networking, CubeSat Space Protocol, Debugging with Logic Analyzer & Oscilloscope.

Software Tools: Altium Design, STM32cubeIDE, VS Code, GitHub, ESP32-IDF, MPLAB X IDE.

Embedded Platforms: AVR, STM32 (ARM Cortex M3), ESP32, Raspberry Pi, SAMV71.

Soft skills: Time Management, Teamwork, Problem solving, Critical Thinking, Research Skills.

EXPERIENCE

Embedded Systems Intern

Student Research Lab – Ho Chi Minh University of Technology | 2025

SAMV71 UART Bootloader

(March 2026 – May 2026)

Description: Designed and implemented a production-style UART bootloader on Microchip **SAMV71**, enabling reliable in-field firmware updates. Developed a custom packet-based protocol with CRC protection and a Python host tool for flashing, verification, and application control.

Responsibilities & Achievements:

- Designed a custom bootloader architecture supporting firmware update, flash erase/write, memory readback, and application jump with vector table relocation.
- Implemented a robust UART communication protocol with packet framing, CRC32 validation, ACK/NACK flow control.

CubeSat Space Protocol (CSP) over CAN & KISS UART

(September 2025 – November 2025)

Description: Engineered a complex, multi-node embedded networking system leveraging CSP, a **TCP/IP-like protocol**, running concurrently on **Linux and STM32**. Enabled modular, reliable communication across relay and end nodes. Implemented reliable file and command transfer through a relay node.

Responsibilities & Achievements:

- Developed **Linux-side C applications** to transmit commands and files over CSP via KISS UART, integrating **Python** processes for file handling and data streaming.
- Developed **STM32** firmware using **FreeRTOS** supporting **CAN** and **UART** for multi-node networks with relay nodes.
- Configured routing tables and service endpoints for **TCP/IP-like** CSP communication.
- Debugged and resolved timing-sensitive, multi-hop communication issues, ensuring robust and reliable file transfer across **Linux ↔ STM32 (relay) ↔ STM32** under real-time constraints.

High-Speed USB File Transfer over CDC

(August 2025 – September 2025)

Description: Bidirectional **USB CDC** file transfer system with custom chunk-based frames; data buffered in STM32 SDRAM, **Python** on laptop handles chunking and file reassembly.

Responsibilities & Achievements:

- Developed **STM32** firmware in **FreeRTOS** and **Python scripts** on laptop for chunk-based file transfer (SOF, SEQ, LENGTH, DATA, EOF).
- Buffered chunks in STM32 **SDRAM via FMC** to reconstruct files; supported sending files back to laptop for reassembly.
- Achieved stable high-throughput bidirectional transfer in lab tests.

Embedded TCP/IP over USB ECM

(June 2025 – August 2025)

Description: Implemented STM32 as a virtual Ethernet adapter on host PC, integrating FreeRTOS+TCP stack based on TCP/IP for lightweight embedded networking with a minimal HTTP server.

Responsibilities & Achievements:

- Enabled **STM32** to be recognized as a **virtual Ethernet adapter** on Linux host PC.
- Integrated **TCP/IP-based FreeRTOS+TCP stack** with lightweight **networking** tasks, including a minimal **HTTP server**.
- Validated end-to-end stack functionality in Windows simulator (ping, TCP echo client, HTTP requests).

USB Composite (HID and CDC)

(April 2025 – June 2025)

Description: Developed a custom USB composite device combining HID and CDC classes on STM32 without using CubeMX/IOC, enabling simultaneous HID and CDC operation.

Responsibilities & Achievements:

- **Manually merged HID and CDC classes** (descriptors, configuration, driver code) on STM32.
- Configured endpoints for concurrent HID + CDC functionality.
- Verified reliable string transmission via HID and interactive console over CDC.

ACADEMIC PROJECTS

Handheld Calculator

- Version 1 (**Raspberry Pi**): Built a calculator receiving keypad input and displaying results on a laptop; implemented arithmetic and equation-solving algorithms in C with **multithreading**.
- Version 2 (**STM32**): Ported the system to STM32 using **FreeRTOS tasks** for parallel execution of **Halley and Secant root-finding methods**, achieving faster computation.

ESP32 BLE Sensor

- Developed BLE application on ESP32 using **ESP-IDF** to read DHT11 temperature and humidity sensor and send data to a mobile app.
- Implemented BLE advertising and **notification payloads**, **GPIO wake-up interrupts**, and **power state management** for low-power operation.

Alarm Clock with Temperature & Humidity (STM32)

- Developed an alarm clock reading DHT11 & DS3231, displaying information on TFT screen; users **set time/alarm** via rotary encoder.
- Programmed STM32 peripherals (GPIO, I2C), implemented buzzer alarm logic, and **stored configuration in Flash**.
- **Designed custom PCB** for compact, stable system operation.

ACTIVITIES

Ho Chi Minh University of Technology Student Research Lab – PAY IT FORWARD

Active member

- Familiar with ARM Microcontroller with peripherals (UART, I2C,...).
- Trained basic C programming for second-year students mentored career paths in Engineering for 30 students and prepared assessments for new members.

EDUCATION

GPA: 7.66

TOEIC: 710

Relevant Coursework: Embedded Systems Design, Embedded Programming (Implemented I2C slave using STM32 LL libraries), Microprocessors (Implemented UART and SPI in Assembly), Applied Electronics.